

Allegro Hand V5 – Tactile Sensor

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Contents

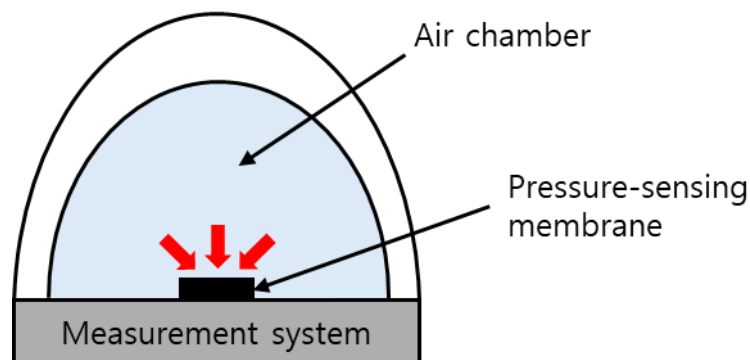
Copyright & Trademark Notice.....	i
Overview.....	1
Principle of Sensor Operation	1
Performance Test.....	2
Precision Test	2
Drift Test	3
Grasping Test	4
Technical Support.....	8

Overview

This tactile sensor—a 360-degree omnidirectional pressure-sensitive tactile sensor—has been developed into a robotic finger with stiffness and shape similar to that of a human finger. It is capable of flexibly handling objects ranging from rigid to deformable soft materials. Additionally, this sensor solves the previous signal distortion issues that occurred depending on the direction of object grasping when applied to 3D-shaped robotic fingers, offering excellent performance and reliability, along with stable characteristics.

Principle of Sensor Operation

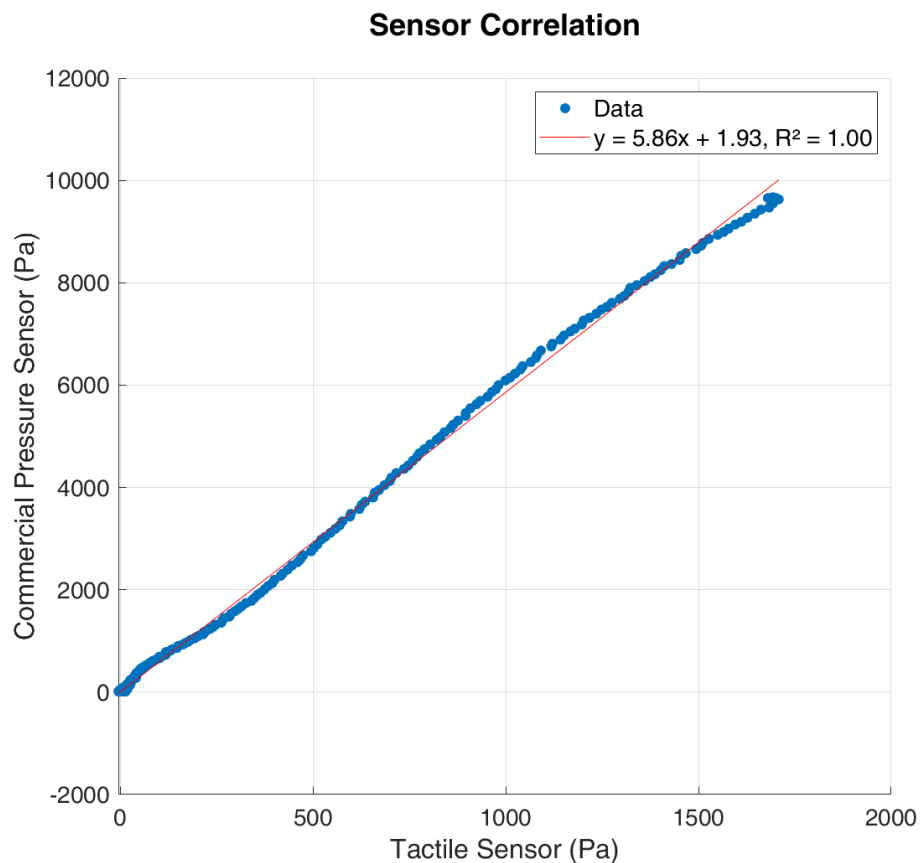
This tactile sensor measures the pressure applied to the sensor surface by detecting pressure changes within a sealed air chamber through a pressure-sensing membrane (piezoelectric) installed at the bottom of the sensor.



Performance Test

Precision Test

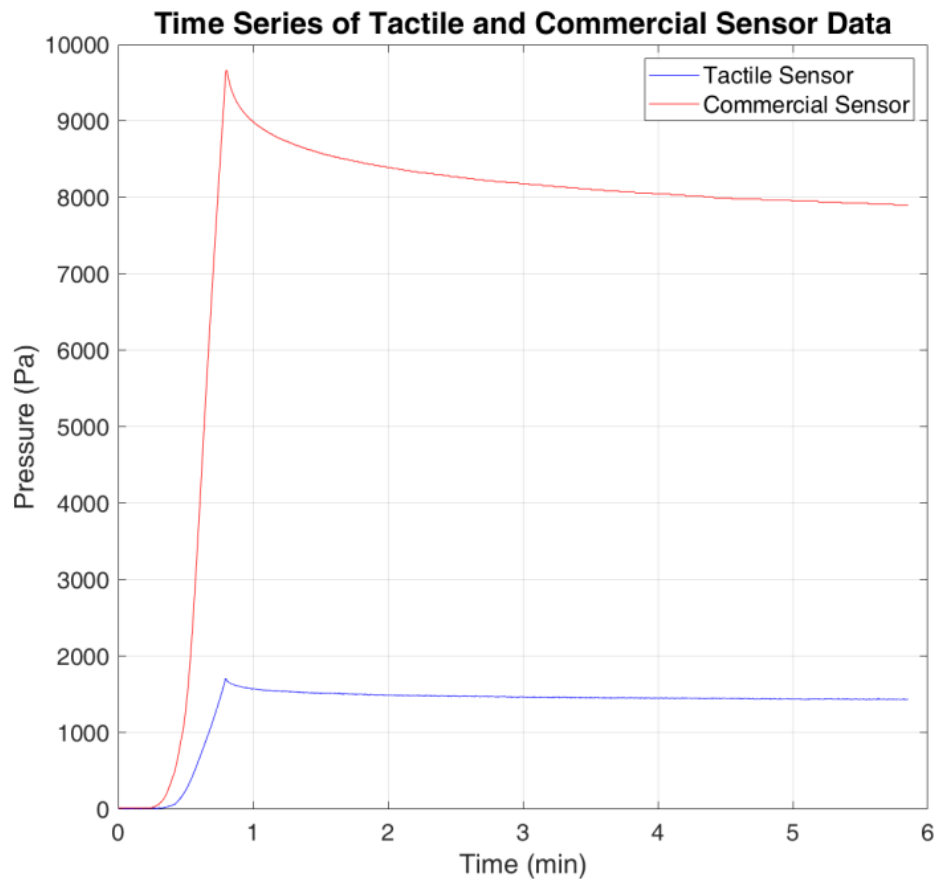
The tactile sensor was connected to a compression and tension tester, and the interaction with a commercial pressure measurement tester was checked. Within the defined pressure application range, a pressure of 10 kPa was applied to the tactile sensor, and the responses from both the pressure measurement tester and the tactile sensor were recorded for each pressure value.



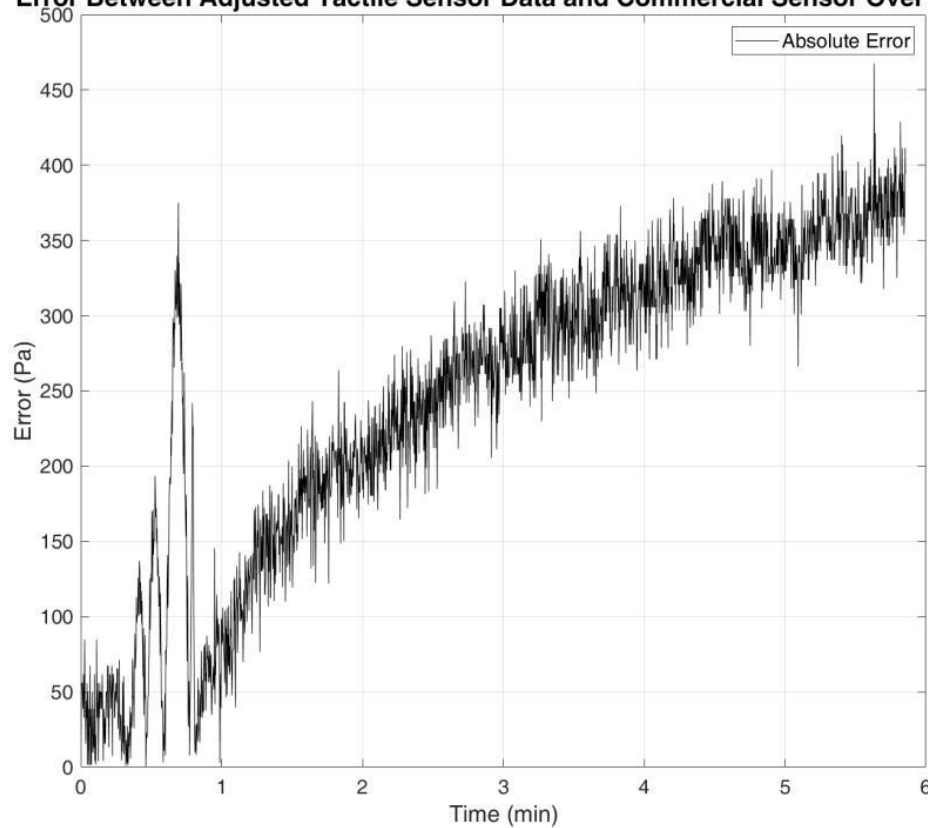
The tactile sensor detected pressure with an accuracy of approximately within 500 Pa in a pressure range of up to 10 kPa, and based on the pressure sensor chip used, it was capable of measuring with a resolution of around 20 Pa.

Drift Test

After applying approximately 5 kPa of pressure to the tactile sensor, it was observed that the sensor maintained a change within 500 Pa compared to a commercial pressure sensor over a certain period of time. This result suggests that the sensor possesses high resilience and stability with respect to the initial pressure. The gradually decreasing pressure values over time can be observed through a graph, which is due to the relaxation, a physical characteristic of the soft material used in the sensor. As the material gradually attempts to return to its original state over time, the measured pressure values gradually decreased.



Error Between Adjusted Tactile Sensor Data and Commercial Sensor Over Time

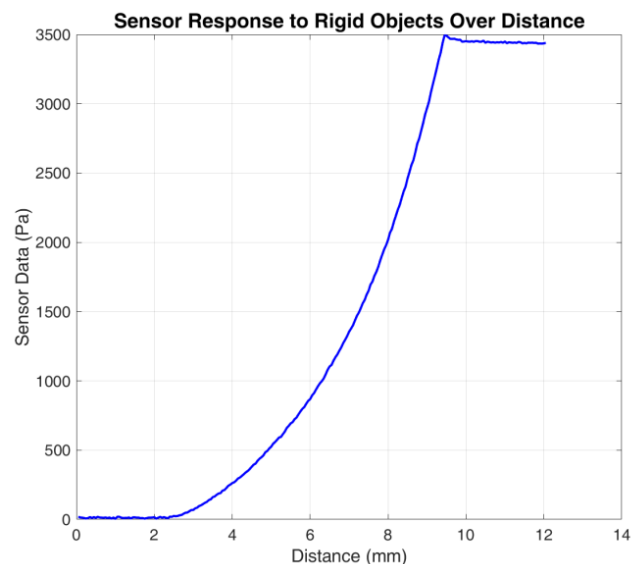
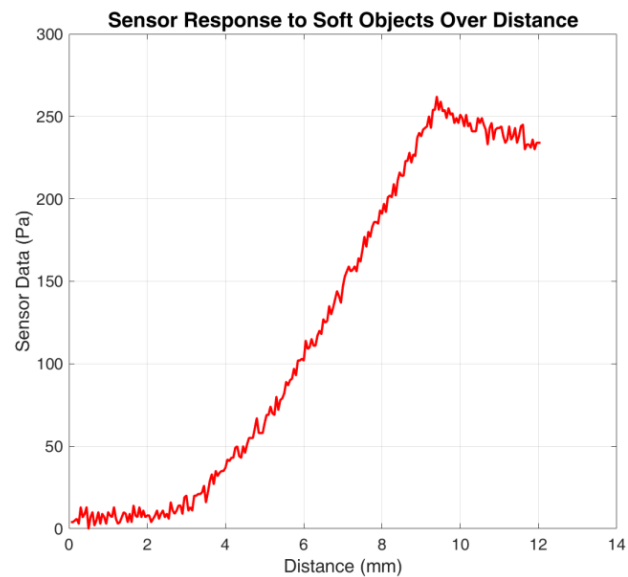


Grasping Test

Tactile sensors were attached to the Allegro Hand to analyze the pressure changes when grasping objects with various material properties. This experiment holds significant value in evaluating the delicate object recognition and manipulation abilities of robotic technology. Specifically, the study aimed to explain how the tactile sensor detects and responds to the characteristics of different objects by comparing and analyzing the pressure changes when grasping hard and soft objects.

First, the pressure changes when grasping soft and hard objects were observed. Soft objects generally have high compressibility, so when the robotic fingers

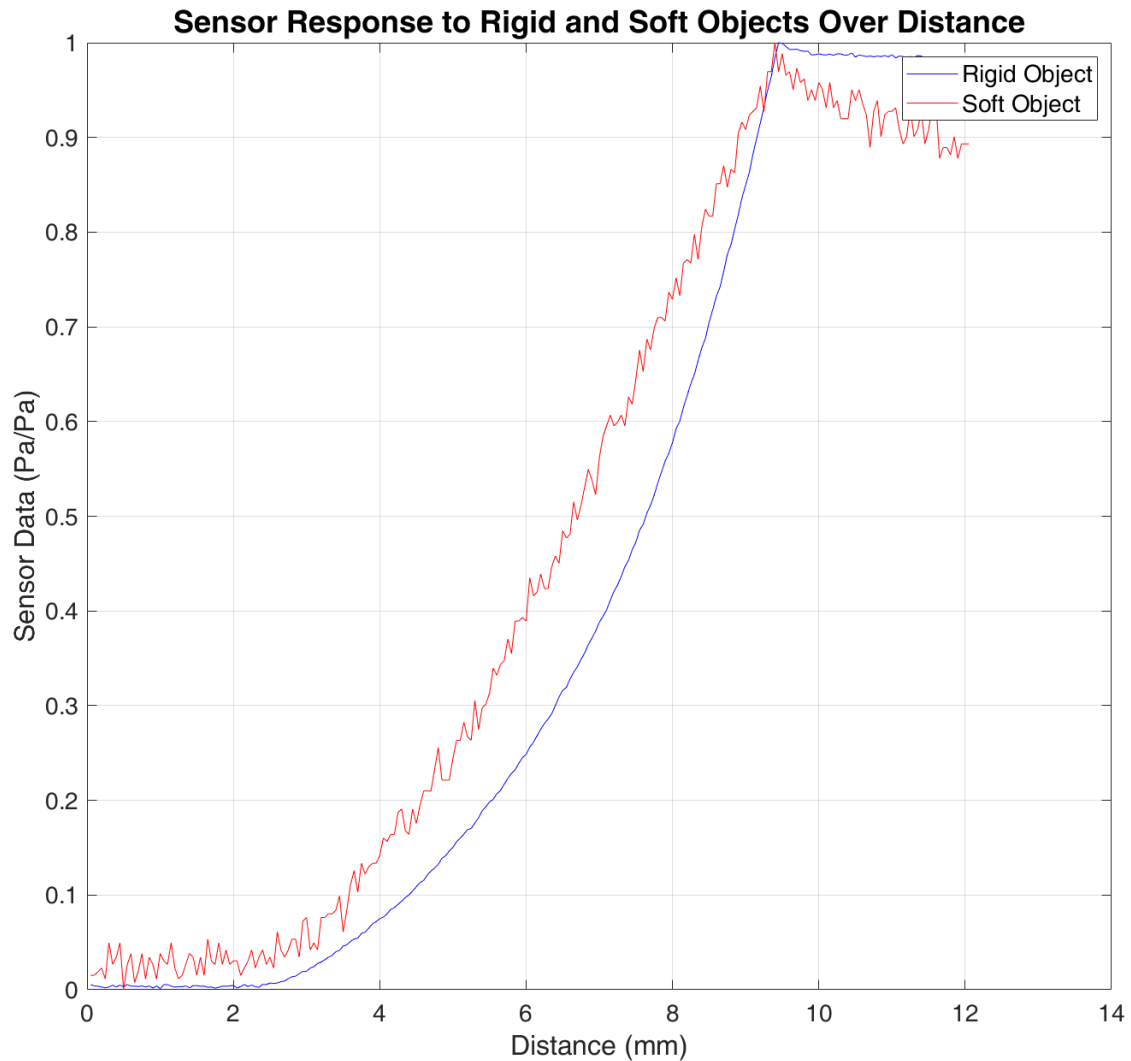
press on them, significant deformation occurs. However, despite this deformation, the pressure change is relatively small because the material of the soft object disperses the pressure. In contrast, hard objects exhibit minimal deformation, leading to a rapid increase in pressure even with small changes. This characteristic is related to the significant increase in internal stress when a hard object is compressed by the robotic fingers.



When analyzing the rate of pressure change according to distance, the pressure for hard objects increases rapidly. This indicates that they are highly sensitive to even small changes in distance due to their elastic limit. On the other hand, soft

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objects, with their greater elasticity, exhibit a relatively gradual pressure increase even when the robotic fingers press deeply. This difference becomes an important factor in allowing the robot to infer the stiffness of the object during manipulation.



The above figure is a normalized graph of the pressure data for two objects. This method allows for a clearer comparison of the differences in pressure responses based on the stiffness of the objects. In the normalized graph, it can be observed that the pressure on the soft object decreases toward the end. This occurs due to the relaxation phenomenon, where the soft object tends to return to its original state after being compressed for a certain period of time. This phenomenon

provides important information for determining whether the robot needs to apply continuous force while holding the object, or whether it should adjust the force after a certain period of time.

Technical Support

The newest version of all documents found in the Allegro Hand website at

<https://www.allegrohand.com/>

A forum is also available on the website where users can share problems and solutions with each other and with WONIK ROBOTICS.

For any further questions please contact us.

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